Ahmad's Week 2

Gen AI Engineering Roadmap

Embeddings · RAG · Vector DBs · PDF Chatbot · Parallel Tracks

Days 8–14 | Built specifically for a MERN developer going all-in on AI Engineering

7 Days	2 Parallel Tracks	2 Projects	1 Deployed App
daily schedule	main + codebasics	PDF chatbot + RAG	live by Sunday

Main Track — Gen AI Engineering

90% of your daily time · Build real AI systems every day

Day 8

Embeddings — how text becomes numbers

The concept that unlocks everything in AI

Morning 1.5hrs

Install + understand embeddings conceptually

Watch: 'Embeddings explained' — Andrej Karpathy YouTube (15 min). Read: platform.openai.com/docs/guides/embeddings first section.

```
pip install sentence-transformers numpy chromadb
```

Evening 1.5hrs

Write your first embedding call + cosine similarity

Compare text meaning with numbers. Try Urdu vs English sentences.

```
from sentence_transformers import SentenceTransformer model =
SentenceTransformer('all-MiniLM-L6-v2') texts = ['I love Python', 'Python is great', 'I eat
biryani'] embeddings = model.encode(texts)
```

Night 30min

Experiment + GitHub commit

Compare: 'AI Engineer' vs 'ML Engineer'. Push: 'Day 8 — embeddings and cosine similarity'

Today's win: You understand embeddings and can compare text meaning in code. Foundation of every RAG system.

Day 9

ChromaDB — your first vector database

Store and retrieve embeddings locally

Morning 1.5hrs

Setup ChromaDB + store your first vectors

Create a collection, add documents with embeddings, query by semantic similarity.

```
import chromadb client = chromadb.PersistentClient(path='./chroma_db') collection =
client.get_or_create_collection('knowledge_base')
```

Evening 1.5hrs	**Persistent DB + add your own facts** Add 10 facts about yourself — skills, projects, experience. Query with 5 different questions. Delete a doc and re-query.
Night 30min	**Read ChromaDB docs + push** Read: docs.trychroma.com — Getting Started. Push to GitHub.

Today's win: Working local vector database. You can store and retrieve meaning, not just text.

Day 10

RAG from scratch — no LangChain

Build the pipeline manually so you actually understand it

Morning 1.5hrs	**Understand + build RAG manually** 4 steps: INGEST chunks → EMBED → STORE in vector DB → QUERY → RETRIEVE → GENERATE. Build WITHOUT LangChain first.

```
def ask(query): chunks = retrieve(query) # get relevant docs return generate(query, chunks) # LLM answers using context
```

Evening 1.5hrs	**Add chunking — split long text properly** Chunk a text file, ingest it, query it. Try chunk sizes 100 vs 300 words. See how answer quality changes.

```
def chunk_text(text, chunk_size=200, overlap=50): words = text.split() return [' '.join(words[i:i+chunk_size]) for i in range(0, len(words), chunk_size-overlap)]
```

Night 30min	**Push + read** Read: python.langchain.com/docs/tutorials/rag — just read tonight, don't code yet.

Today's win: You built RAG from scratch. You understand every piece. Most people just use LangChain and have no idea what's inside.

Day 11

LangChain — now speed it up

Use the framework now that you understand what it's doing

Morning 1.5hrs	**Install LangChain + rebuild RAG in 20 lines** Compare to your Day 10 code — same result, 5x less code. Now you appreciate what LangChain does.

```
pip install langchain langchain-community langchain-groq chromadb
```

Evening 1.5hrs	**LangChain document loaders + text splitters** Load a text file → split → store → query. Install pypdf and try loading a real PDF file.

```
from langchain.text_splitter import RecursiveCharacterTextSplitter splitter = RecursiveCharacterTextSplitter(chunk_size=500, chunk_overlap=50)
```

Night 30min	**Push + DeepLearning.AI course** Start: deeplearning.ai/short-courses — 'LangChain for LLM Application Development'. Do first 2 lessons tonight.

Today's win: You can build RAG both manually and with LangChain. You know when to use which.

Day 12

PDF chatbot — your week 2 project

Upload any PDF and chat with it — full stack

Morning 2hrs

Build the Python backend — FastAPI + RAG

FastAPI endpoint: POST /upload processes PDF → chunks → embeds → stores. POST /ask queries and returns answer.

```
pip install fastapi uvicorn python-multipart pypdf langchain-groq uvicorn main:app --reload
```

Evening 2hrs

Build the React frontend

Use AI freely for the UI. Focus on: PDF upload area, progress indicator showing chunk count, chat interface (same as week 1), disable input until PDF uploaded.

Night 30min

Test everything locally

Upload your own CV as a PDF. Ask: 'What is Ahmad's experience?' Verify the answer is accurate.

Today's win: Working PDF chatbot on localhost. Upload any PDF — CV, contract, textbook — and chat with it.

Day 13

Polish + deploy live

Nothing counts until it is live on the internet

Morning 2hrs

Add one impressive feature

Pick one: show which page the answer came from / auto-generate 3 questions after upload / 'summarise this PDF' button.

Evening 2hrs

Deploy — Railway + Vercel

pip freeze > requirements.txt. Add Procfile. Deploy FastAPI to Railway (add GROQ_API_KEY). Deploy React to Vercel. Test live end to end.

```
web: uvicorn main:app --host 0.0.0.0 --port $PORT
```

Night 30min

Write README + record demo

Proper README with screenshots + live link. Record 30-second demo: upload PDF, ask 3 questions, show streaming answers.

Today's win: Live deployed PDF chatbot. This is a real product you can show clients and charge for.

Day 14

Week 2 wrap-up + LinkedIn + prep week 3

Document, share, and plan ahead

Morning 1.5hrs

LinkedIn week 2 post + screen recording

Hook: 'I just built something that would have cost $50k from an agency 3 years ago.' Attach demo recording. Tag @Groq.

Evening 1.5hrs

Preview week 3 — AI agents

Read: Anthropic blog 'What are AI agents'. Watch: LangGraph intro by Harrison Chase (20 min YouTube).

```
pip install langgraph tavily-python crewai
```

Night 30min	**Week 2 checklist** Embeddings ✓ | ChromaDB ✓ | RAG scratch ✓ | LangChain ✓ | PDF chatbot deployed ✓ | LinkedIn + GitHub updated ✓

Today's win: Week 2 complete. You went from zero RAG knowledge to a deployed PDF chatbot. Week 3 — agents. This is where it gets insane.

Parallel Track — Codebasics Foundation

30 min/day only · Never blocks your main track

Day 8–9: Advanced Python patterns
- Watch Codebasics Python videos 17–22: Inheritance, Generators, Iterators
- These patterns appear in LangChain internals — good to recognise them
- Do NOT spend more than 30 min — just watch, don't build exercises yet

Day 10–11: NLP fundamentals — why embeddings work
- Watch Codebasics NLP playlist: Tokenization, stemming, lemmatization
- Watch: Word2Vec explained — this is the ancestor of modern embeddings
- Count vectorizer + TF-IDF — understand WHY semantic search is better

Day 12–13: List comprehensions + Decorators + Lambda
- Codebasics Python videos 23–27: Decorators, Multithreading concepts
- Practice: rewrite 3 of your existing Python functions using list comprehensions
- 30 min only — these come naturally as you build more

Day 14: Soft skills — LinkedIn engagement strategy
- Comment meaningfully on 5 AI-related LinkedIn posts today
- Follow: Andrej Karpathy, Harrison Chase, Jerry Liu on X and LinkedIn
- Subscribe: The Rundown AI newsletter (therundown.ai) — 5 min daily AI news

---

Week 2 Resources

Everything you need — free unless marked

Type	Resource	What to do
Free course	DeepLearning.AI — LangChain for LLM Apps	Do every notebook. Andrew Ng + Harrison Chase
Free course	DeepLearning.AI — Building RAG Systems	Hands-on. Very practical. Essential this week.
Docs	python.langchain.com/docs/tutorials/rag	Official RAG tutorial. Read fully on Day 11 night.
YouTube	Sam Witteveen — RAG tutorials	Best practical RAG coding channel. Watch his pla
Tool (free)	ChromaDB — docs.trychroma.com	Local vector DB. Getting Started section.
Tool (free)	sentence-transformers library	pip install sentence-transformers — free embeddi
Platform	Railway.app	Free tier. Deploy your FastAPI backend here.
Platform	Vercel.com	Free tier. Deploy React frontend. Connect GitHub
YouTube	Andrej Karpathy — Intro to LLMs	1-hour talk. Best conceptual foundation. Watch Da

Week 2 GitHub commit checklist

Push something every single day

Day	File to push	Commit message
Day 8	day8_embeddings.py	Day 8 — embeddings and cosine similarity
Day 9	day9_chromadb.py	Day 9 — ChromaDB local vector database
Day 10	day10_rag_scratch.py	Day 10 — RAG pipeline built from scratch
Day 11	day11_langchain_rag.py	Day 11 — RAG with LangChain and document loaders
Day 12	pdf_chatbot/backend/main.py	Day 12 — PDF chatbot FastAPI backend
Day 13	pdf_chatbot/ (full)	Day 13 — PDF chatbot deployed live [link]
Day 14	README.md update	Week 2 complete — PDF chatbot shipped

> **Week 2 in one sentence:** You go from calling AI APIs to building a complete AI system that understands documents — and you deploy it live for the world to use.